

Investigating the Impact of Virtual Avatars and Owner Gender on Virtual Partner Selection in Avatar-based Interactions: A Study Design

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ABSTRACT

As real-life social interactions transition to virtual environments using Virtual/Augmented Reality (VR/AR) technologies, understanding how gender representation in virtual avatars affects choices and behaviors becomes increasingly relevant. In this paper, we investigate the combined impact of avatars' gender representation and their owner's gender on virtual partner selection in physical, intellectual, social, and romantic scenarios. We introduce our preliminary research plan, outlining both the study design and system development. The research will contribute to our understanding of social dynamics and gender effects in avatar-mediated interactions.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

As Virtual/Augmented Reality (VR/AR) technologies continue to advance, virtual environments have evolved into spaces where individuals connect with others forging meaningful connections, friendships, and even romantic relationships [3]. In this dynamic landscape of virtual environments, the interplay between digital personas and human interactions has become a subject of increasing significance [5]. As individuals immerse themselves in virtual spaces, the alignment or mismatch between virtual avatars and their owners, e.g., in their gender and appearance, holds implications for the nature of social interactions within these digital realms [1, 2].

In this paper, we present our early-stage research plan aiming to delve into the complex relationship between a virtual avatar's gender and the gender of its owner. The research focuses specifically on their combined impact on virtual partner selection while addressing the following research questions:

- RQ1.** Does the *visually* presented gender of a virtual avatar influence a user's preference in selecting the avatar as an interaction/collaboration partner?
- RQ2.** Does the *vocally* presented gender of a virtual avatar influence a user's preference in selecting the avatar as an interaction/collaboration partner?
- RQ3.** Does the avatar owner's gender impact a user's preference in selecting the avatar as an interaction/collaboration partner?

While previous research has explored the impact of avatar gender on user cognition [4], the simultaneous examination of both the avatar's gender and the gender of its owner in partner selection contexts introduces a novel dimension to the investigation. The

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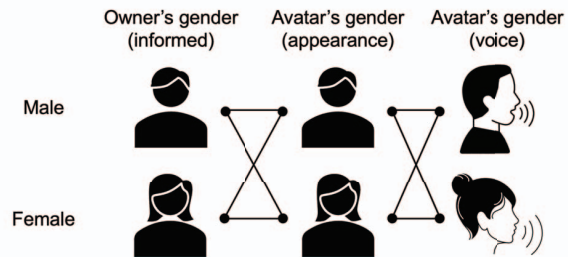


Figure 1: The gender combinations of the owner and avatar in the planned study.

exploration of virtual partner choice within the context of avatar and owner gender not only contributes to our understanding of online social dynamics but also has implications for fields ranging from human-computer interaction to the psychology of digital identity.

2 PLANNED STUDY

To tackle the research questions above, we plan to conduct a human-subject study with different avatar and owner gender conditions.

Study Design The study incorporates three gender factors: (1) the informed avatar owner's gender, (2) the visually represented avatar's gender (appearance), and (3) the vocally presented avatar's gender (voice). Despite the variety of gender categories, we use only two levels for each factor, i.e., male and female, which make a $2 \times 2 \times 2$ study design (Figure 1). To reduce the individual differences in gender perception/preference, we plan to use a within-subject design where participants experience animated virtual avatars in different gender settings. To mitigate the potential influence of order effects, a Latin square design is employed to balance the sequence of the scenes. The research ethics application is currently under review by the University of Calgary Research Ethics Board.

Scenarios In the study, participants will experience animated virtual avatars that introduce themselves, and be asked to report their preference for each avatar in four partner selection scenarios described below. These scenarios will be given as short passages.

- **Physical Task *Dynamic Tennis Duo*:** Pair up with a tennis partner for an exhilarating contest! Choose your teammate, strategize together, and share in the triumph as your combined skills make a winning impact on the court. It's not just a tennis match; it's a collaboration for victory!
- **Intellectual/Academic Task *Collaborative Final Exam*:** For the final exam, Prof. Weasley allows students to form a union with a classmate. Although they still take the test individually in person, the final grade is determined by averaging the scores of both union members. Success is not merely in individual scores but in the collaborative journey of learning together.
- **Communication/Social Task *Startup Duo Challenge*:** In this entrepreneurial challenge, participants pair up to craft a compelling pitch for their startup idea. This dynamic duo will present their innovation to a panel of "investors" or potential

sponsors in the Shark Tank setting. The emphasis is not just on individual skills but on effective collaboration and persuasive communication. After the pitch, the investors may ask a few brief questions to test the teams' ability to respond on the spot.

- **Romantic Task** *Tinderverse: Virtual Date*: With Valentine's Day just around the corner, envision an ideal date at the beach or a charming coffee shop. Grab your partner and dive into a virtual romantic experience.

Avatar/Owner Personas There are eight imaginary individuals, each assuming the role of the avatar owner in a respective scene. To rule out potential confounding effects other than the avatar and owner gender factors, we designed eight imaginary personas to share an identical set of attributes. They are 20-year-old second-year college students majoring in International Relations. Additionally, they are members of the campus cycling club, regularly commuting to school on their bicycles. They actively participate in the International Culture Committee, organizing cultural exchange events.

Procedure Upon arrival and after obtaining the participant's informed consent, they will complete a demographic questionnaire. Participants will experience a total of eight scenes, corresponding to the combinations of the three gender factors. Before the actual experiment, the experimenter will verbally inform the owner's gender of the avatar that the participant will encounter, emphasizing the variability of the avatar's owner across scenes. Once the experiment starts, participants will encounter an avatar positioned in the lab room through an Augmented Reality (AR) headset. Avatars will introduce themselves, covering details such as name, major, GPA, hobbies, sports activities, and social engagements based on the pre-defined persona profile. Following the avatar's one-minute presentation, participants will remove the headset and assess the likelihood of choosing the recently encountered avatar as a partner for the four task scenarios. This process will be repeated eight times to ensure participants experience all conditional combinations. Subsequently, they will provide open-ended feedback, allowing for more details of their overall experience and partner decisions.

Measures and Hypotheses Participants will provide essential demographic information, including details on sexual orientation and gender, aiding in the contextualization of their preferences. The primary measure involves participants assessing their likelihood of choosing each encountered avatar as a partner for the four distinct tasks, using a 7-point Likert scale (1: low likelihood to 7: high likelihood). Although we do not provide concrete hypotheses for all combinations of study conditions, we generally anticipate significant differences in participants' partner selection preferences concerning the gender factors and the task scenarios:

- H1.** The avatar's (visual/vocal) gender will influence the partner selection preference for the avatar.
- H2.** The owner's gender will influence the partner selection preference for the avatar.
- H3.** The impacts of gender factors will vary depending on the task scenarios.

3 SIMULATED AVATAR SYSTEM DEVELOPMENT

For the planned study, we created two gender avatars—a female character and a male character. We used Character Creator 4¹ for the characters, and Mixamo² animations and iClone 8³ for the body gestures. Both are depicted as young Caucasians wearing identical clothes and sharing the same animation (Figure 2). To ensure equivalence in competence and physical activities, we standardized the

¹Reallusion Character Creator 4: <https://www.reallusion.com/character-creator/> (Accessed: 2023-12-16)

²Adobe Mixamo: <https://www.mixamo.com/> (Accessed: 2023-12-16)

³Reallusion iClone 8: <https://www.reallusion.com/iclone/> (Accessed: 2023-12-16)



Figure 2: The male and female virtual characters for the study.

height and overall size of the characters. This intentional balance prompts users to evaluate the avatars based on virtual representation and gender information rather than inherent biases. These characters will be presented to the study participants together with the pre-defined owner's profile. In fact, these avatars will not be controlled by human users (owners), but autonomous agents. In the scene, the avatar will talk based on the scripts as if the owner is introducing themselves through the avatar.

The system was built on the Unity Game Engine (version 2022.3.4f1), specifically configured as an AR application compatible with the Meta Quest 3 headset. It comprises a total of eight distinct scenes, each showcasing one of the two avatars talking within a one-minute timeframe. Their audios were created using a text-to-speech software that generates human-like dialog audios, Murf.AI⁴. We used young adult voice models for the female and male characters.

4 CONCLUSION

In this research, we aim to investigate the impacts of the gender presentation of the avatar and the owner's gender on virtual partner selection. This paper presents our preliminary study plan and the developed avatar system for the study. The insights gained from this research will contribute valuable knowledge to our understanding of human perceptions of virtual avatars, enhancing virtual communication and relationships.

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⁴Murf.AI: <https://murf.ai/>